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R V College of Engineering

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

IV Semester BE Regular/Supplementary Examinations June/July -2025.
Course : Statistics For Data Analytics-IM241AT

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	In what way is a stem and leaf diagram better than a histogram?	02	1	2
1.2	A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A U B U C U D, and each element of S is equally likely to happen, then $P(A' \cap B)$ =__	02	2	2
1.3	For a continuous random variable, the probability that it takes an exact value is ________.	01	2	2
1.4	A negative binomial distribution counts the number of trials needed to get 3 successes with p=0.4. What is the probability that the 3^{rd} success occurs on the 5^{th} trial?	02	2	2
1.5	A random variable X is uniformly distributed over the interval [2,6]. Find the probability that X lies between 3 and 5.	02	2	2
1.6	For a normal distribution with $\mu=70$, $\sigma=8$, find $P(62<X<78)$	02	2	2
1.7	If a population has a mean μ and standard deviation σ, the standard error of the mean is given by σ/sqrt{n}, where n is the __________.	01	2	2
1.8	From Minitab output: Coefficients: Intercept = 2.5, Slope = 1.8, If X = 6, what is the predicted value of Y?	02	2	2

1.9	What does a correlation coefficient of -0.85 indicate?	02	3	2
1.10	In linear regression the error term is distributed as _____ distribution (include its parameters also).	02	3	2
1.11	The test statistic in a test for population variance based on a given sample of size n is __________ (also state what the terms are).	02	4	2

Part B

Question No	Question	M	CO	BT
2a	The no of defects when 25 parts were analyzed were recorded as: 15, 18, 20, 9, 11, 8, 16, 14, 13, 15, 17, 21, 7, 9, 11, 8, 18, 12, 6, 19, 15, 22, 19, 12, 10 a. Draw the box plot for the above data b. Plot the stem and leaf diagram and comment on the data	08	1	3
2b	Suppose it is known that 1% of the population suffers from a particular disease. A blood test has a 97% chance of a positive blood test (identifying the disease) for diseased individuals, but also has a 6% chance of falsely indicating that a healthy person has the disease. (i) What is the probability that a person will have a positive blood test? (ii) If your blood test is positive, what is the chance that you have the disease?	08	1	4
3a	The probability mass function of a discrete random variable X is defined as: a) Find the expected value E(X) b) Find the probability that X takes a value greater than 3.	08	2	3
3b	The random variable X has a binomial distribution with n = 10 and p = 0.01. Determine the following probabilities. a) P (X = 5) b) $P(X \leq 2)$ c) $P(X \geq 9)$	08	2	2

	d) $P(3 \le X < 5)$			
OR				
4a	If a random variable X has its cumulative distribution as $F(x) = 0, -\infty < x < 1,$ $0.4, 1 \le x < 3,$ $0.5, 3 \le x < 8,$ $1.0, 8 \le x < \infty$ Determine the following: (i) P(X at most 3) (ii) E(X) (iii) P(X between 1 and 2, both inclusive) (iv) P(X at least 2) (v) the median of the distribution	10	1	3
4b	A carton of washers contains 5% whose variability in thickness around the circumference is unacceptable. i) What is the probability that none of the unacceptable washers is found in a sample of 10 washers? ii) What is the probability that at least one unacceptable washer is found in a sample of 20 washers?	06	2	3
5a	Describe the characteristics and applications of the Normal distribution.	04	2	2
5b	What are the key differences between population and sample? Why is sampling important in statistics?	06	2	2
5c	What is the Central Limit Theorem (CLT)? State its significance in statistical inference.	06	2	3
OR				
6a	An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find a. the probability that a bulb burns between 778 and 834 hours. b. Probability that a bulb has a life of at least 1000 hours c. What percent of bulbs have their life within $\pm 2\sigma$ from their own mean life, where σ is the standard deviation?	08	2	3
6b	The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours.	08	2	3

	a) What is the probability that a laser fails before 6000 hours? b) What is the life in hours that 95% of the lasers exceed? c) If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours?			
7a	Regression methods were used to analyze the data from a study investigating the relationship between roadway surface temperature (x) and pavement deflection (y). Summary quantities were n = 20. $\sum yi=12.75$, $\sum yi^2=8.86$, $\sum xi=1478$, $\sum xi^2=143{,}215.8$, and $\sum xiyi=1083.67$ a. Calculate the least squares estimates of the slope and intercept. Graph the regression line. Estimate σ^2. b. Use the equation of the fitted line to predict what pavement deflection would be observed when the surface temperature is 85°F. c. What is the mean pavement deflection when the surface temperature is 90°F? d. What change in mean pavement deflection would be expected for a 1°F change in surface temperature?	12	3	3
7b	Define the correlation coefficient and its range.	04	3	2
	OR			
8a	The summarized data on 10 observations are given below. a. Estimate β1 and βo assuming positive correlation between X and Y. b. Obtain an estimate of the underlying variance. c. What is the proportion of variability explained by the regression model? d. Assuming normal distribution of the errors, how will a residual plot of errors vs predicted values appear in this case? (Show an indicative plot). $\sum_i (y_i - \hat{y}_i)^2 = 3{\cdot}267,\ \sum_i y_i = 63{\cdot}3,\ \sum_i y_i^2 = 423{\cdot}49,\ \sum_i x_i = 139,\ \sum_i x_i^2 = 2239$	10	3	3
8b	What are the assumptions of simple linear regression? How do you validate them using residual plots?	06	3	2
9a	Explain the steps involved in conducting a hypothesis testing.	08	4	2
9b	a) What are the hypothesis statements and the statistic to be used in a two-sided test of variance of a population? What are the assumptions to use this statistic and what is the rejection region	08	4	4

	for this test? b) State the probability of a t-random variable with 25 degrees of freedom taking on any value between 2.06 and 3.45 c) What is the value x such that a chi-square random variable with 15 degrees of freedom takes on at the most a value of x with 95% likelihood?			
OR				
10a	When is a z-test used for population mean inference?	04	4	2
10b	Cloud seeding has been studied for many decades as a weather modification. The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate are as follows: 18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, and 31.6. (a) construct a 95 % one-sided lower bound on the mean diameter. (b) construct a 90 % two-sided confidence interval on the mean diameter. What is the width of this interval? (c) If the width of the interval in part (b) were to be halved, how much should be the sample size? Use trial and error method if needed.	12	4	4

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R V College of Engineering

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Mysuru Road Bengaluru - 560 059

IV Semester BE Regular/Supplementary Examinations June/July - 2025.
Course : Operations Research-IM244AI

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of normal distribution table is permitted.

Part A

Question No	Question	M	CO	BT
1.1	Operations research is a .... approach to problem solving for executives.	01	1	2
1.2	Any two isoprofit or isocost lines for a given linear programming problem are .... to each other.	01	2	1
1.3	The constraints may be in the form of .... or	02	2	1
1.4	Define sensitivity analysis in linear programming.	02	1	1
1.5	What does the shadow price in duality represent?	02	2	2
1.6	What is the goal in a transportation problem?	01	2	1
1.7	What is the cost matrix in assignment problems?	01	2	2
1.8	What are the steps of the Hungarian method?	02	3	2
1.9	What is the full form of CPM?	01	1	1
1.10	What is a critical path?	01	2	2

1.11	What are the advantages of crashing a network?	02	3	2
1.12	Define saddle point in a game matrix.	02	2	1
1.13	When is a game solved using the arithmetic method?	02	4	2

Part B

Question No	Question	M	CO	BT
2a	With an example, explain a work-scheduling problem in linear programming.	06	2	3
2b	Explain the concept of feasible region and optimal solution in a linear programming problem using an example.	06	2	3
2c	Define Operations Research and mention any four of its applications.	04	1	1
3a	Use penalty (Big-M) method to solve the following LP problem. Minimize Z = 5x1 + 3x2 subject to the constraints (i) 2x1 + 4x2 ≤ 12, (ii) 2x1 + 2x2 = 10, (iii) 5x1 + 2x2 ≥ 10 and x1, x2 ≥ 0.	08	2	3
3b	Formulate the **dual** of the following problem and interpret the dual variables economically: Maximize Z=6x+8y s.t. 2x+3y≤30, x+2y≤20, x,y≥0	08	2	3
OR				
4a	Write the standard format of simplex method for the following LP problem. Maximize Z = 3x1 + 5x2 + 4x3 subject to the constraints (i) 2x1 + 3x2 ≤ 8, (ii) 2x2 + 5x3 ≤ 10, (iii) 3x1 + 2x2 + 4x3 ≤ 15 and x1, x2, x3 ≥ 0	06	2	4
4b	Conduct **sensitivity analysis** for the objective coefficients of the LP: Maximize Z=2x+3y s.t. x+y≤4 x+2y≤6 x,y≥0 after obtaining optimal solution.	10	2	4
5a	A company wants to assign 4 salesmen (S1, S2, S3, S4) to 4 different sales regions (R1, R2, R3, R4). The expected profits (in ₹‘000s) from	10	3	3

each salesman in each region are given in the table below. Assign the salesmen to the regions such that the total profit is maximized.

	R1	**R2**	**R3**	**R4**
S1	25	40	35	20
S2	30	45	40	25
S3	20	35	30	50
S4	10	25	20	30

5b	Compare the three methods of finding initial basic feasible solution: North West Corner, Least Cost, Vogel's Method.	06	2	4

OR

6a	Discuss degeneracy in a transportation problem. How is it identified and resolved?	06	3	2
6b	Explain the general structure and assumptions of a Transportation Problem.	06	1	2
6c	Explain any two methods to find the initial basic feasible solution.	04	2	2
7a	A project consists of the following activities with the duration in days: (table below)	10	2	4

Activity	A	B	C	D	E	F	G	H	I
Precedence	-	A	A	B,C	A	D,E	C	F,G	H
Duration	10	4	12	10	9	8	7	9	6

i. Draw the network of the above project
ii. Identify the critical path and determine the project duration
iii. Calculate independent float, free float for all the activities.

7b	An assembly is to be made from two parts X and Y. Both parts must be turned on a lathe. Y must be polished whereas X need not be polished. The sequence of activities, together with their predecessors, is given below. (table below) Draw a network diagram of activities for the project	06	3	4

Activity	Description	Predecessor Activity
A	Open work order	–
B	Get material for X	A
C	Get material for Y	A
D	Turn X on lathe	B
E	Turn Y on lathe	B, C
F	Polish Y	E
G	Assemble X and Y	D, F
H	Pack	G

OR

<table>
<tr><td>8a</td><td>‘PERT takes care of uncertain durations.’ How far is this statement correct? Explain with reasons.</td><td>08</td><td>3</td><td>4</td></tr>
<tr><td>8b</td><td>Explain the following in the context of project management:
(i) Activity variance
(ii) Project variance</td><td>08</td><td>3</td><td>2</td></tr>
<tr><td>9a</td><td>Explain: Minimax and Maximin principle used in the theory of games.</td><td>06</td><td>4</td><td>2</td></tr>
<tr><td>9b</td><td>By applying dominance rules, reduce the given payoff matrix and identify the optimal pure strategies and value of the game.
<table><tr><td></td><td>B1</td><td>B2</td><td>B3</td><td>B4</td></tr><tr><td>A1</td><td>1</td><td>2</td><td>1</td><td>3</td></tr><tr><td>A2</td><td>0</td><td>-1</td><td>0</td><td>2</td></tr><tr><td>A3</td><td>2</td><td>3</td><td>1</td><td>4</td></tr></table></td><td>10</td><td>4</td><td>4</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>10a</td><td>Obtain the optimal strategies for both persons and the value of the game for two-person zero-sum game whose payoff matrix is as follows:
<table><tr><td></td><td colspan="2">Player B</td></tr><tr><td>Player A</td><td>B_1</td><td>B_2</td></tr><tr><td>A_1</td><td>1</td><td>−3</td></tr><tr><td>A_2</td><td>3</td><td>5</td></tr><tr><td>A_3</td><td>−1</td><td>6</td></tr><tr><td>A_4</td><td>4</td><td>1</td></tr><tr><td>A_5</td><td>2</td><td>2</td></tr><tr><td>A_6</td><td>−5</td><td>0</td></tr></table></td><td>10</td><td>4</td><td>3</td></tr>
<tr><td>10b</td><td>Game theory provides a systematic quantitative approach for analysing competitive situations in which the competitors make use of logical processes and techniques in order to determine an optimal strategy for winning. Comment.</td><td>06</td><td>4</td><td>5</td></tr>
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R V College of Engineering

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

IV Semester Regular/Supplementary Examinations June/July- 2025
Course : CAD/CAM & Robotics-IM343AI

Time : 3 Hours

Maximum Marks : 100

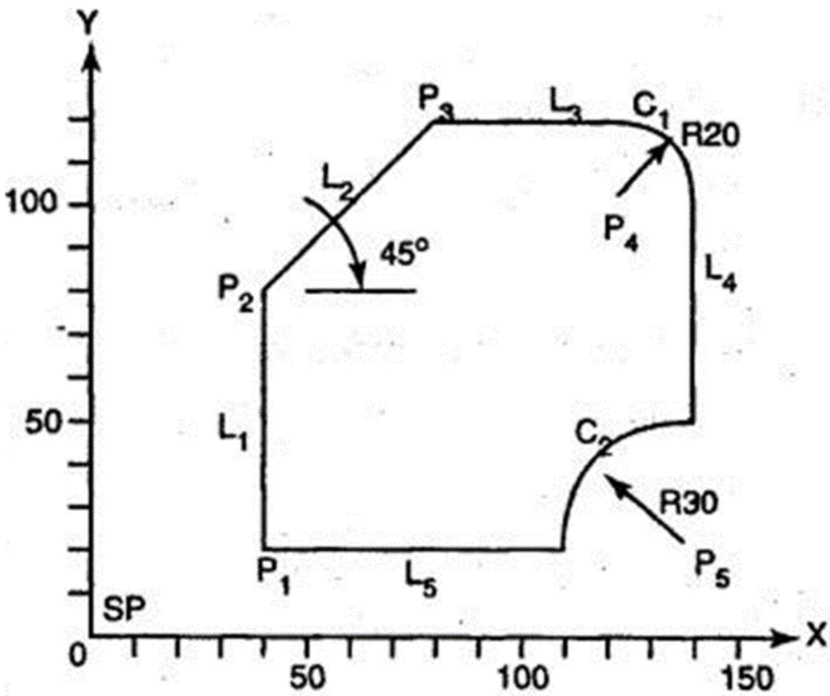
Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Why is conceptual design an important phase in the Shigley model?	02	1	1
1.2	What is the purpose of a line drawing algorithm?	02	1	1
1.3	What is a part program in NC machining?	02	2	2
1.4	What is tool offset in CNC machining?	02	2	2
1.5	What is mechanization?	02	2	2
1.6	List two motivating factors for robot usage in industries	02	3	1
1.7	The power source of a robot can be electric, hydraulic, or ______.	01	3	1
1.8	What is meant by payload in robotics?	02	4	2
1.9	Give two examples of exteroceptive sensors	02	4	1
1.10	What is lead-through programming?	02	4	1
1.11	A __________ is a device at the end of a robot arm, designed to interact with the environment	01	4	1

Part B

<table>
<tr><th>Question No</th><th>Question</th><th>M</th><th>CO</th><th>BT</th></tr>
<tr><td>2a</td><td>Use Bresenham’s Line Algorithm to rasterize a line from (10, 15) to (20, 25). Show all intermediate steps.</td><td>10</td><td>1</td><td>4</td></tr>
<tr><td>2b</td><td>Discuss the various applications of CAD in different industries such as mechanical, civil, and electrical engineering.</td><td>06</td><td>1</td><td>3</td></tr>
<tr><td>3a</td><td>Analyze the effect of incorrect tool offset values in CNC milling programs.</td><td>06</td><td>2</td><td>1</td></tr>
<tr><td>3b</td><td>Compare APT programming with manual NC programming in terms of flexibility, accuracy, and complexity.</td><td>10</td><td>2</td><td>3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>4a</td><td>Prepare a program for milling the given component by milling process using APT.
<table>
<tr><td>Post Processor</td><td>Machine Mill, 1</td></tr>
<tr><td>Feed rate</td><td>60 mm / min</td></tr>
<tr><td>Spindle speed</td><td>500 rpm</td></tr>
<tr><td>Cutter diameter</td><td>5 mm</td></tr>
<tr><td>Tolerance</td><td>± 0.02 mm</td></tr>
<tr><td>Thickness</td><td>10 mm</td></tr>
</table>


</td><td>16</td><td>2</td><td>4</td></tr>
</table>

5a	Explain the social issues associated with automation	08	2	1
5b	Explain reasons for adopting automation in industries.	08	3	3
OR				
6a	Explain how automation contributes to sustainable manufacturing	08	3	4
6b	Explain the different types of automation with examples.	08	2	2
7a	Discuss the selection criteria for industrial robots in manufacturing applications.	08	3	2
7b	Explain the robot wrist and end-of-arm tooling (EOAT) with examples	08	3	2
OR				
8a	A family-owned metal fabrication workshop is evaluating whether to automate its manual cutting and welding operations. However, the owner is hesitant. Analyze the reasons why automation might not be suitable for such a workshop. Provide alternatives.	10	4	4
8b	Explain the classification of robots based on configuration	06	3	1
9a	Describe in detail the role of sensors in robotics and how they improve robot performance.	08	4	3
9b	How do exteroceptive sensors support robotic adaptability in dynamic environments?	08	4	2
OR				
10a	A bakery needs a robot to pick and place soft pastries into boxes. Recommend an end effector and justify it based on hygiene, delicacy, and cycle time.	10	4	5
10b	Explain on-line robot programming and its advantages	06	4	1

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R V College of Engineering

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Mysuru Road Bengaluru - 560 059

IV Semester BE Regular/Supplementary Examinations June/July - 2025.
Industrial Engineering and Management
Course : Marketing Management-IM345AT

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Differentiate between push and pull marketing.	02	2	2
1.2	Mention any four characteristics of a good content.	02	1	1
1.3	What is the primary goal of social media marketing?	02	1	1
1.4	What is the primary objective of Search Engine Optimization (SEO)?	02	1	2
1.5	How bounce rate affects website performance assessment.	02	2	2
1.6	Why is segmentation important in e-mail marketing campaigns?	02	2	2
1.7	Why is it important to integrate SEO in website development?	02	4	2
1.8	What is the role of geo-targeting in mobile marketing?	02	2	2
1.9	Why is the 'Satisfaction' component important in the AIDAS model?	02	2	1
1.10	What is meant by the “end-to-end customer experience” in digital analytics?	02	3	2

Part B

Question No	Question	M	CO	BT
2a	Discuss how website analysis contributes to enhanced customer experience and conversion optimization.	08	2	3
2b	Explain how content research tools help marketers discover new opportunities and gain a competitive edge.	08	2	3

3a	Analyse how businesses can utilize Facebook and Instagram for different stages of a marketing funnel.	08	2	3
3b	Develop an SEO content plan for an e-commerce startup targeting eco-friendly household products.	08	2	3
OR				
4a	llustrate how social media analytics contribute to creating audience profiles.	08	3	3
4b	Construct a targeted marketing plan using SEO tools like Google Analytics and SEMrush for a B2B company.	08	3	4
5a	Design a simple marketing plan for an e-learning platform using insights gathered from web analytics tools.	08	3	4
5b	Analyze a failed e-mail marketing campaign and identify areas where marketing research could have improved outcomes.	08	3	3
OR				
6a	Evaluate the effectiveness of responsive email design in maintaining a competitive edge in digital marketing.	08	4	4
6b	Critically evaluate the contribution of Google Tag Manager in optimizing marketing channels for global companies.	08	4	3
7a	Demonstrate how tools like Google Analytics and Hotjar can be used to develop insights into user behavior and improve marketing decisions.	08	3	3
7b	Formulate a mobile-based marketing plan for a local food delivery startup to reach a targeted student segment.	08	3	3
OR				
8a	Evaluate the effectiveness of mobile marketing in facing market changes due to rapid technological advancements.	08	2	4
8b	Create a strategy to use new digital channels like mobile apps, chatbots, and social media integration to strengthen a brand's global online presence.	08	4	4
9a	What are the key performance indicators (KPIs) used to evaluate website optimization, and how do they contribute to increased conversions?	08	4	3
9b	Analyze how market segmentation can be refined using digital analytics data.	08	3	3
OR				
10a	Design a basic digital dashboard that reflects holistic marketing performance.	08	2	4
10b	Analyze how a company can use the AIDAS model to structure content on its product landing page.	08	3	3